CE4052/CZ4052 Cloud Computing

Semester 2 Examination 2022-2023 — Model Answers

Question 2 uses the corrected graph provided by the user:

- $1 \to \{2, 4\}$
- $2 \to \{4\}$
- $3 \to \{1, 2\}$
- $4 \to \{3\}$

Question 1

1(a) Two advantages of Hadoop / MapReduce for distributed systems (10 marks)

1. Scalability on commodity clusters MapReduce splits a job into many independent map tasks and reduce tasks. This allows a very large data set to be processed across many low-cost machines.

- In **word count**, each mapper processes a different input shard and emits $(w, 1)$ pairs.
- The reducers aggregate all occurrences of the same word.
- If the input doubles in size, we can usually scale out by adding more worker nodes instead of replacing one machine with a much larger one.

This is a major advantage in distributed systems because web-scale workloads often grow beyond the limits of a single server.

2. Built-in fault tolerance and simplified programming model The programmer only writes:

- $\text{map}(k, v) \to \text{list}(k_2, v_2)$
- $\text{reduce}(k_2, \text{list}(v_2)) \to \text{list}(k_3, v_3)$

The framework handles:

- task scheduling,
- data partitioning,
- shuffle/sort,
- failure recovery,
- re-execution of failed tasks,
- moving computation close to data.

Example: in **PageRank**, each iteration can be written as a MapReduce job:

- Map: page j emits contribution r_j/d_j to each outgoing neighbor.
- Reduce: sum incoming contributions to produce the next PageRank value.

If one worker crashes during a large iteration, Hadoop simply re-runs that task on another node. The user does not need to implement distributed coordination manually.

Summary Two strong advantages are:

1. **horizontal scalability** using many commodity machines,
2. **automatic fault tolerance plus a simple programming abstraction**.

These make Hadoop/MapReduce effective for applications such as word count, inverted indexing, log analysis, and PageRank.

1(b) 3-stage Close network for $N = 16$, $n = 4$, $k = 6$, $m = 4$ (15 marks)

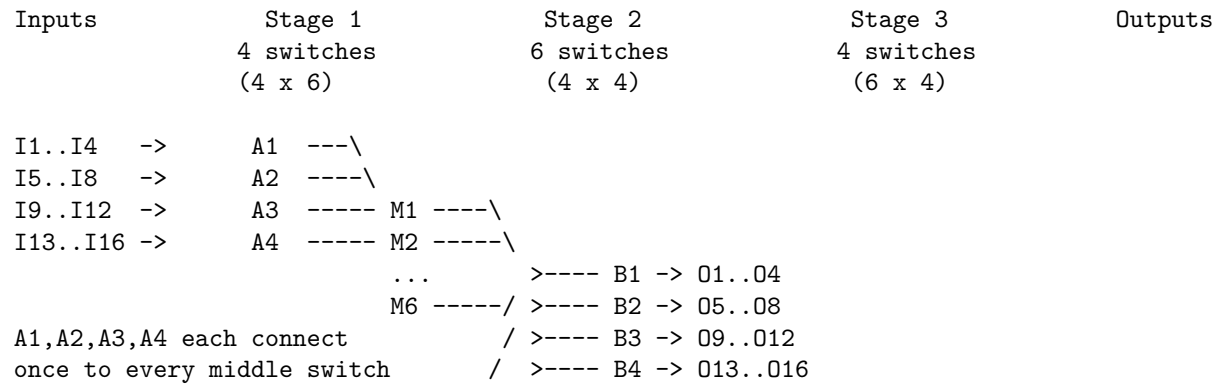
For a 3-stage Close network $C(n, m, r)$ with total inputs $N = rn$:

$$r = N/n = 16/4 = 4$$

So the topology is:

- **Stage 1:** $r = 4$ switches, each of size $n \times k = 4 \times 6$
- **Stage 2:** $k = 6$ switches, each of size $m \times m = 4 \times 4$
- **Stage 3:** $r = 4$ switches, each of size $k \times n = 6 \times 4$

One valid drawing (ASCII)



Each first-stage switch has 4 inputs and 6 outputs, one to each middle switch. Each middle switch has 4 inputs from the 4 first-stage switches and 4 outputs to the 4 third-stage switches.

Can we always add a new connection without rearrangement? No.

A 3-stage Close network is **strict-sense nonblocking** only if:

$$m \geq 2n - 1$$

Here:

$$2n - 1 = 2(4) - 1 = 7$$

but the question gives:

$$m = 4 < 7$$

So this network is **not strict-sense nonblocking**. Therefore, we **cannot always add a new connection without rearrangement**.

What property does it have? A Close network is **rearrangeably nonblocking** if:

$$m \geq n$$

Here $m = 4$ and $n = 4$, so this condition holds. Hence the network **can realize any valid permutation, but may require rearranging some existing connections first**.

Conclusion

- **Without rearrangement:** not always possible.
- **With rearrangement:** yes, because $m = n$ makes it rearrangeably nonblocking.

Question 2

Given graph

The corrected graph is:

- $1 \to \{2, 4\}$
- $2 \to \{4\}$
- $3 \to \{1, 2\}$
- $4 \to \{3\}$

Using the standard PageRank balance equations with teleportation probability 0:

$$r_1 = \frac{r_3}{2}$$
$$r_2 = \frac{r_1}{2} + \frac{r_3}{2}$$
$$r_3 = r_4$$
$$r_4 = \frac{r_1}{2} + r_2$$

with normalization:

$$r_1 + r_2 + r_3 + r_4 = 1$$

2(a) Compute the PageRank and rank the webpages (10 marks)

From the graph,

$$r_1 = \frac{r_3}{2}$$
$$r_2 = \frac{r_1}{2} + \frac{r_3}{2}$$
$$r_3 = r_4$$

Substitute $r_1 = r_3/2$ into the equation for r_2:

$$r_2 = \frac{1}{2}\left(\frac{r_3}{2}\right) + \frac{r_3}{2} = \frac{r_3}{4} + \frac{r_3}{2} = \frac{3r_3}{4}$$

Using normalization:

$$r_1 + r_2 + r_3 + r_4 = 1$$
$$\frac{r_3}{2} + \frac{3r_3}{4} + r_3 + r_3 = 1$$
$$\left(\frac{2}{4} + \frac{3}{4} + \frac{4}{4} + \frac{4}{4}\right) r_3 = 1$$
$$\frac{13}{4} r_3 = 1$$
$$r_3 = \frac{4}{13}$$

Hence:

$$r_4 = \frac{4}{13}$$

$$r_1 = \frac{r_3}{2} = \frac{2}{13}$$
$$r_2 = \frac{3r_3}{4} = \frac{3}{13}$$

Therefore the PageRank vector is:

$$\boxed{(r_1, r_2, r_3, r_4) = \left(\frac{2}{13}, \frac{3}{13}, \frac{4}{13}, \frac{4}{13}\right)}$$

Ranking

$$3 = 4 > 2 > 1$$

So webpages 3 and 4 tie for the highest PageRank, followed by 2, then 1.

2(b) Teleportation probability changes to 0.1 (10 marks)

With teleportation probability 0.1, the random surfer:

- follows a hyperlink with probability 0.9,
- teleports uniformly to one of the 4 pages with probability 0.1.

A quick approximation using the result in 2(a) is:

$$r' \approx 0.9r + 0.1 \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

Using

$$r = \begin{bmatrix} 2/13 \\ 3/13 \\ 4/13 \\ 4/13 \end{bmatrix}$$

we get:

$$r'_1 \approx 0.9\left(\frac{2}{13}\right) + 0.025 = 0.1385 + 0.025 = 0.1635$$

$$r'_2 \approx 0.9\left(\frac{3}{13}\right) + 0.025 = 0.2077 + 0.025 = 0.2327$$

$$r'_3 = r'_4 \approx 0.9\left(\frac{4}{13}\right) + 0.025 = 0.2769 + 0.025 = 0.3019$$

So the approximation is:

$$\boxed{r' \approx (0.1635,\ 0.2327,\ 0.3019,\ 0.3019)}$$

Comment on quality This approximation is good because the teleportation probability is still small. The exact solution is slightly different:

$$(r_1, r_2, r_3, r_4) = (0.1604,\ 0.2325,\ 0.3008,\ 0.3064)$$

So the approximation preserves the correct broad ordering, with pages 3 and 4 remaining the most important and page 1 remaining the least important.

Question 3

3(a) EDF with non-work-conserving VMs (10 marks)

Given five VMs:

- VM1: $(1, 8, 0)$
- VM2: $(1, 5, 0)$
- VM3: $(2, 6, 0)$
- VM4: $(2, 10, 0)$
- VM5: $(2, 15, 0)$

Their utilizations are:

$$u_1 = \frac{1}{8}, \quad u_2 = \frac{1}{5}, \quad u_3 = \frac{2}{6} = \frac{1}{3}, \quad u_4 = \frac{2}{10} = \frac{1}{5}, \quad u_5 = \frac{2}{15}$$

Total utilization:

$$U = \frac{1}{8} + \frac{1}{5} + \frac{1}{3} + \frac{1}{5} + \frac{2}{15}$$

Using denominator 120:

$$U = \frac{15 + 24 + 40 + 24 + 16}{120} = \frac{119}{120} < 1$$

So the set is **schedulable**.

Additional VM(s) that can still be admitted Remaining spare utilization is:

$$1 - \frac{119}{120} = \frac{1}{120}$$

Therefore any additional non-work-conserving VM with specification $(x, y, 0)$ is admissible if:

$$\frac{x}{y} \leq \frac{1}{120}$$

Equivalently:

$$y \geq 120x$$

So all possible additional single-VM specifications are:

- $(1, 120, 0), (1, 121, 0), (1, 122, 0), ...$
- $(2, 240, 0), (2, 241, 0), (2, 242, 0), ...$
- $(3, 360, 0), (3, 361, 0), ...$
- in general, **any** $(x, y, 0)$ with $y \geq 120x$.

If multiple new VMs are admitted together, their total additional utilization must satisfy:

$$\sum_i \frac{x_i}{y_i} \leq \frac{1}{120}$$

That is the complete admission condition.

3(b) All VMs become work-conserving (10 marks)

Now the specifications are:

- VM1: $(1, 8, 1)$
- VM2: $(1, 5, 1)$
- VM3: $(2, 6, 1)$
- VM4: $(2, 10, 1)$
- VM5: $(2, 15, 1)$

The guaranteed utilizations are unchanged, so over the hyperperiod the baseline reserved CPU quanta are proportional to their utilizations.

The hyperperiod is:

$$\text{lcm}(8, 5, 6, 10, 15) = 120$$

Guaranteed CPU quanta in one hyperperiod:

VM	Utilization	CPU quantum in 120 time units
VM1	1/8	15
VM2	1/5	24
VM3	1/3	40
VM4	1/5	24
VM5	2/15	16

The total guaranteed CPU is:

$$120 \cdot \frac{119}{120} = 119$$

So there is only **1 spare quantum** in each hyperperiod that can be used as extra time in work-conserving mode.

Which gets the most CPU quantum? VM3 has the largest guaranteed share:

$$40 \text{ quanta}$$

So **VM3 gets the most CPU quantum**.

Which gets the least CPU quantum? VM1 has the smallest guaranteed share:

$$15 \text{ quanta}$$

So **VM1 gets the least CPU quantum**.

Explanation Work-conserving mode only changes what happens to the **single leftover quantum**. Since there is only 1 spare unit per 120-unit hyperperiod, this does not change the overall ordering of the VMs by CPU share.

Hence:

Most: VM3, Least: VM1

Question 4

4(a) CAP theorem for distributed crowdsourcing systems, using ReCAPTCHA and Google Instant Search (10 marks)

The CAP theorem states that when a **network partition** occurs, a distributed system cannot simultaneously guarantee all three of:

- **Consistency (C):** all users see the same latest state,
- **Availability (A):** every request gets a response,
- **Partition tolerance (P):** the system continues operating despite communication failure between nodes.

In practice, a large internet-scale system must tolerate partitions, so the real design tradeoff is usually between **consistency** and **availability**.

Applying CAP to crowdsourcing systems A distributed crowdsourcing system coordinates many geographically distributed users who submit responses, labels, clicks, or human-verification results. During a partition, the system can either:

1. **Prefer consistency (CP):** block or delay some requests to avoid conflicting global state.
2. **Prefer availability (AP):** accept responses everywhere and reconcile conflicts later.

ReCAPTCHA ReCAPTCHA is a good crowdsourcing example because users solve challenges, and the collected human responses help classify content or improve models.

A ReCAPTCHA service usually favors **availability + partition tolerance** for challenge delivery:

- the service should still present a challenge quickly,
- it should still accept human responses from many regions,
- temporary replica inconsistency is acceptable because answers can be aggregated or reconciled later.

So ReCAPTCHA is typically **AP-leaning** in its data collection path.

However, for security-sensitive decisions such as fraud detection or abuse scoring, some internal components may use **CP-like behavior** to avoid accepting contradictory or duplicated trusted state.

Google Instant Search Google Instant Search is also strongly **AP-leaning**:

- users expect extremely low latency,
- the service must remain responsive worldwide,
- slightly stale rankings or suggestions are acceptable for a short time.

If a partition happens, it is better to return slightly outdated suggestions than to block all search suggestions. That means the system prioritizes **availability and partition tolerance**, while allowing weaker consistency.

Conclusion For distributed crowdsourcing systems:

- If the main goal is fast response and large-scale participation, the design tends to be **AP**.
- If the main goal is strict correctness of shared global state such as payments, one-time task assignment, or exact accounting, the design tends to be more **CP**.

Using the examples:

- **ReCAPTCHA:** mostly AP for challenge serving and answer collection.

- **Google Instant Search:** strongly AP because responsiveness is more important than perfectly synchronized state.

4(b) Consistent hashing event trace and load balance (15 marks)

The OCR of the question contains typographical errors, but the pattern matches the other exam papers. I use:

- $M = 11$
- server replica hashes:

$$f(ip) = ip \bmod M$$

$$g(ip) = ((ip + 51) \cdot 150) \bmod M$$

- data replica hashes:

$$f(d) = d \bmod M$$

$$g(d) = (d + 51) \bmod M$$

Step 1: Server positions on the ring Since $150 \bmod 11 = 7$:

Server s_1 with $ip = 1$

$$f(1) = 1$$

$$g(1) = ((1 + 51) \cdot 150) \bmod 11 = (52 \cdot 150) \bmod 11$$

$$52 \bmod 11 = 8, \quad 8 \cdot 7 = 56 \equiv 1 \pmod{11}$$

So s_1 is at positions **1 and 1**.

Server s_2 with $ip = 40$

$$f(40) = 40 \bmod 11 = 7$$

$$g(40) = ((40 + 51) \cdot 150) \bmod 11 = (91 \cdot 150) \bmod 11$$

$$91 \bmod 11 = 3, \quad 3 \cdot 7 = 21 \equiv 10 \pmod{11}$$

So s_2 is at positions **7 and 10**.

Server s_3 with $ip = 80$

$$f(80) = 80 \bmod 11 = 3$$

$$g(80) = ((80 + 51) \cdot 150) \bmod 11 = (131 \cdot 150) \bmod 11$$

$$131 \bmod 11 = 10, \quad 10 \cdot 7 = 70 \equiv 4 \pmod{11}$$

So s_3 is at positions **3 and 4**.

Final active servers The event sequence includes: start s_1, start s_2, later start s_3, then s_3 fails.

So at the **end**, the active ring contains only:

- s_1 at position 1
- s_2 at positions 7 and 10

Data item	$f(d) = d \bmod 11$	$g(d) = (d + 51) \bmod 11$

Step 2: Final data hash positions For each data item, compute two replicas:

Data item	$f(d) = d \bmod 11$	$g(d) = (d + 51) \bmod 11$
$d = 10$	10	6
$d = 20$	9	5
$d = 30$	8	4
$d = 40$	7	3
$d = 60$	5	1
$d = 70$	4	0
$d = 80$	3	10

Step 3: Assign each replica clockwise on the final ring Final ring positions:

- position 1: s_1
- position 7: s_2
- position 10: s_2

Therefore:

- positions 0 and 1 map to s_1
- positions 2 through 7 map to s_2
- positions 8 through 10 map to s_2

So the final responsibilities are:

Data item	Replica positions	Responsible server(s)
$d = 10$	10, 6	s_2, s_2
$d = 20$	9, 5	s_2, s_2
$d = 30$	8, 4	s_2, s_2
$d = 40$	7, 3	s_2, s_2
$d = 60$	5, 1	s_2, s_1
$d = 70$	4, 0	s_2, s_1
$d = 80$	3, 10	s_2, s_2

Is the load balanced at the end? No.

At the end:

- s_1 covers only the short arc ending at position 1,
- s_2 covers almost the entire ring using positions 7 and 10,
- s_1 also suffers because both of its virtual replicas collide at the same position 1.

Counting final replicas:

- s_1 stores only **2** replicas
- s_2 stores **12** replicas

So the load is highly skewed toward s_2.

Conclusion The load is **not balanced** at the end because:

1. one server failed (s_3),
2. s_1's two virtual replicas collapse to the same position,
3. the remaining active server positions are unevenly spread on the ring.

4(c) Probability that n random points lie on a semicircle, and effect on consistent hashing (10 marks)

For n points placed independently and uniformly at random on a circle, the probability that **all n points lie within some semicircle** is:

$$\boxed{\Pr(\text{all } n \text{ points lie on a semicircle}) = \frac{n}{2^{n-1}}}$$

Why? A standard argument is:

- Pick the leftmost point of the semicircle boundary among the n points.
- For all points to fit in a semicircle starting at that point, the remaining $n-1$ points must all fall within the next half-circle.
- Each remaining point independently has probability $1/2$ of landing in that semicircle.
- There are n choices for which point is the boundary point.

Hence:

$$n\left(\frac{1}{2}\right)^{n-1} = \frac{n}{2^{n-1}}$$

Effect on consistent hashing load distribution In consistent hashing, server positions are random points on a ring. If many points happen to lie within a semicircle, then:

- the opposite half of the ring contains very few or no servers,
- one or a few servers will own a very large arc,
- many data items mapped into that large arc will concentrate on those servers.

So this probability measures one form of **load imbalance risk**.

As n increases,

$$\frac{n}{2^{n-1}} \to 0$$

very quickly, which means severe one-sided clustering becomes less likely. This is why using **many virtual nodes** improves balance in consistent hashing: more random points on the ring reduce the chance of large empty arcs and smooth out the load distribution.

End of 2023 Model Answers